

BellaBeat Data Workflow

Business Task:

BellaBeat is a company that focuses on smart wearable products for women. The purpose of this study was to analyze data from another smart product brand to find trends in the data that may be useful for BellaBeat. In this study I analyzed sleep and activity data.

Stakeholders:

Urška Sršen: Bellabeat's cofounder and Chief Creative Officer

Sando Mur: Mathematician and Bellabeat's cofounder; key member of the Bellabeat executive team

Datasource:

The dataset used for this analysis was the: "Fitbit Fitness Tracker Data" from Kaggle. The dataset includes key metrics from Fitbit users such as: Daily Activity, Steps, Calories Burned, Sleep, and other health related metrics.

During this analysis I focused on two key files from the dataset:

- Daily_Activity_Merged.csv
- Sleep_Day_Merged.csv

One important thing to note is that the dataset is rather small. It is also not directly from BellaBeat so some discretion is needed.

Tools Used:

Considering the size of the dataset I worked exclusively in Google Sheets. I used formulas, summary tables, data quality flags, pivot tables, and charts to manipulate and transform the data.

Data Cleaning and Processing:

In order to preserve the data after downloading it I separated it into two folders. One I intended to modify and the other I did not. Once I got working I added a "Clean Date" column as well as a new metric called "Minutes Awake In Bed". This metric was calculated by subtracting the amount of time from "TotalTimeinBed" from "TotalMinutesAsleep".

Next I created a new tab called "Daily_Summary". The purpose of this was to create a table that would show each user's averages on a daily basis. The averages include: steps, calories

burned, and distance all as daily averages. I also included two additional columns for “Number of Sleep Days Tracked” and “Activity Data Quality Flag”.

I also created a sleep_summary table. This included: average minutes asleep, average hours asleep, average time in bed, number of sleep days tracked, and a sleep data quality flag.

After creating the daily activity summary and sleep summary tables I then joined them by user_id into a new table called “combined_user_summary”. This made it easy to see the relationship between activity and sleep per user.

Lastly, I segmented users into three different categories based on average steps.

Low: Under 5,000 steps

Moderate: 5,000-9,999 steps

High: Over 10,000 steps

This categorized my data a bit more by providing context on whether certain categories of each group had more sleep, calories burned, etc.

Analysis Summary:

The Activity Summary contained 35 users. Of those 35 users only 24 had sleep data.

The average steps across all users was 6,538. The median steps are 6,104. This suggests that the average user falls into the moderate activity range.

Average sleep is 6.29 hours. The median however is 6.98 hours. This suggests that there are some outliers on the lower side of the sleep average.

After segmenting the users by activity level the results were as follows:

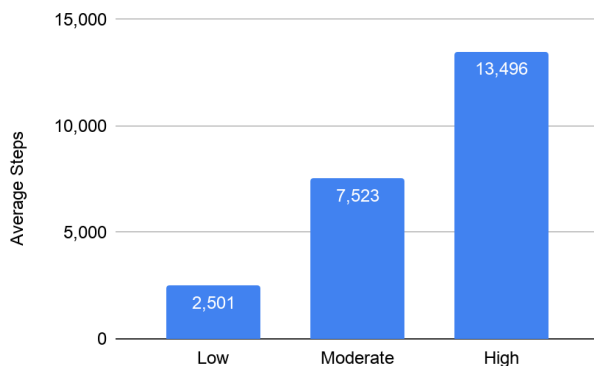
ActivityLevel	UserCount	Avg Steps	Avg Calories	Avg Distance	AvgHour Asleep	AvgMinutes Awake InBed	UsersWithSleep Data	UsersWithoutSleep Data
Low	14	2501	1861	1.72	6.81	63	7	7
Moderate	15	7523	2338	5.34	6.45	41	13	2
High	6	13496	2555	9.93	4.89	12	4	2

This shows that the more active users burned more calories. The sleep data is a bit inconclusive in my opinion due to a lack of enough quality data.

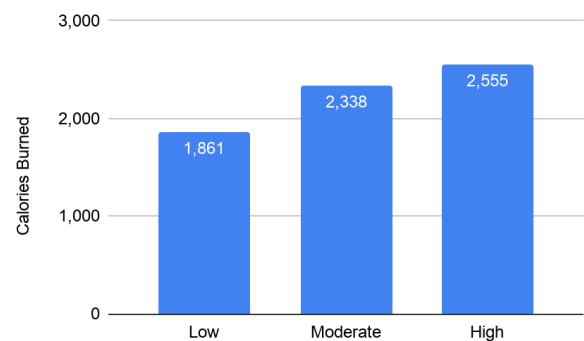
Key Findings:

- Most users fall into the low or moderate categories. 29/35 of the users average under 10,000 steps.
- Calories Burned varies by activity level greatly. Low users averaged 1,861. Moderate averaged 2,338. High averaged 2,555.
- Sleep data is rather incomplete and therefore should be taken with a grain of salt.
- Low activity users spend much more time awake in bed. There is an opportunity to promote being more active in order to resolve this issue.

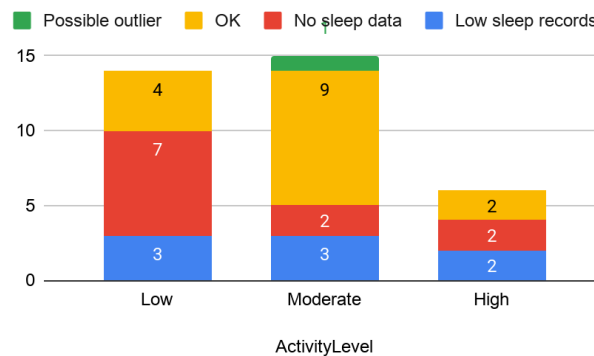
Average Steps By Activity Level



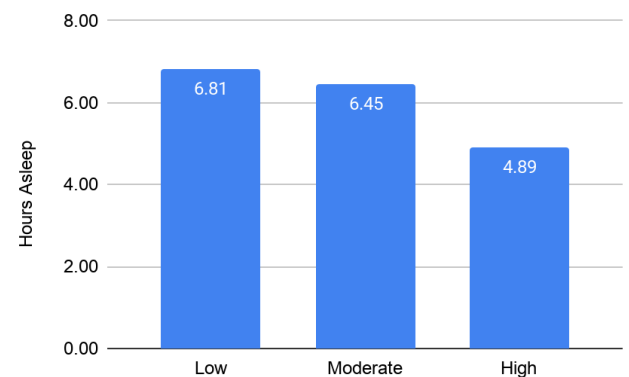
Average Calories Burned By Activity Level



Sleep Data Quality By Activity Level



Average Sleep Hours By Activity Level



Suggestions:

Personalize app messaging/fitness plans by segment for BellaBeat users.

Promote sleep tracking more through reminders, recovery insights, and weekly summaries.

Position BellaBeat subscription marketing to be a personalized recovery, activity, hydration and sleep tracker that delivers real insights and manageable goals by activity level.

Limitations:

This dataset is from a different wearable company, "Fitbit". It also has incomplete sleep records alongside being a smaller dataset. Calories burned is also dependent on device accuracy. These results should be treated directionally rather than exactly. BellaBeat customers also may have different habits.